PESONary Test Cases

MCO1 Monolith Project · GCash FinTech Engineer Specialist Course

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1. Scope

Testing covers the service and controller classes, following the instructor's MCO1 FAQ. Every public method in those classes has at least three documented, unique test cases, labelled Success, Failure or Edge. In total, 37 methods are covered by 114 test cases.

Private helpers (debit, credit, requireActive, findOrThrow and the convertToDTO mappers) are tested through the public methods that call them.

2. How the tests run

- Service tests are unit tests with JUnit 5 and Mockito. Repositories are mocked, so no database or Spring context is involved. Method security annotations do not run in these tests.
- Controller tests are slice tests with @WebMvcTest and MockMvc. The service is mocked with @MockitoBean, and filters are disabled (addFilters = false), so the tests check HTTP status codes, response JSON, validation and the GlobalExceptionHandler mapping.
- Access rules (401 and 403) are checked end to end in Postman, shown in the recorded demo.
- Run all tests with .\mvnw.cmd test on Windows or ./mvnw test on macOS and Linux. The JaCoCo coverage report is written to target/site/jacoco/index.html.
- Test sources: src/test/java/com/pesonary/ewalletapi/service and src/test/java/com/pesonary/ewalletapi/controller.

3. Summary

Class	Test type	Methods covered	Test cases
AccountService	Unit (Mockito)	7	21
CustomerService	Unit (Mockito)	6	18
TransactionService	Unit (Mockito)	5	17
AccountController	Slice (@WebMvcTest + MockMvc)	6	18
AuthController	Slice (@WebMvcTest + MockMvc)	2	7
CustomerController	Slice (@WebMvcTest + MockMvc)	6	18
TransactionController	Slice (@WebMvcTest + MockMvc)	5	15
Total		37	114

Last test run: [to be completed: date and the Tests run / Failures line from mvnw test]

Line coverage (JaCoCo): [to be completed: percentage from target/site/jacoco/index.html]

4. Test cases

Each ID is TC-<class>-<number>. The test method column is the JUnit method name in the test class, so every row can be traced to its code.

4.1 AccountService

Test class: service/AccountServiceTest.java · Unit (Mockito)

ID	Method	Test method	Type	Input / precondition	Expected result
TC-AS-01	createAccount	createAccount_defaultsToActiveBasic	Success	Customer 1 exists; DTO: SAVINGS, ₱1,000.00, no status or tier	Returns status ACTIVE and tier BASIC
TC-AS-02	createAccount	createAccount_unknownCustomer	Failure	Customer 1 does not exist	ResourceNotFoundException "Customer not found"
TC-AS-03	createAccount	createAccount_overBasicLimit	Edge	Opening balance ₱10,000.01 (BASIC cap ₱10,000.00)	BusinessRuleException "Opening balance exceeds BASIC limit"; nothing saved
TC-AS-04	getAllAccounts	getAllAccounts_returnsAll	Success	Repository holds 2 accounts	List of 2 accounts
TC-AS-05	getAllAccounts	getAllAccounts_empty	Edge	Repository is empty	Empty list
TC-AS-06	getAllAccounts	getAllAccounts_mapsCustomerName	Success	1 account owned by customer "Alice"	customerName is "Alice"
TC-AS-07	getPaginatedAccounts	getPaginatedAccounts_returnsContent	Success	page 0, size 5; page holds 1 account	Page content has 1 account
TC-AS-08	getPaginatedAccounts	getPaginatedAccounts_empty	Edge	page 0, size 5; empty page	Empty page
TC-AS-09	getPaginatedAccounts	getPaginatedAccounts_usesGivenPageAndSize	Success	page 2, size 10	Repository queried with PageRequest.of(2, 10)
TC-AS-10	getAccountsByStatus	getAccountsByStatus_active	Success	"ACTIVE"; 1 active account	List of 1 account
TC-AS-11	getAccountsByStatus	getAccountsByStatus_invalidStatus	Failure	"BOGUS"	IllegalArgumentException

ID	Method	Test method	Type	Input / precondition	Expected result
TC-AS-12	getAccountsByStatus	getAccountsByStatus_lowercaseWorks	Edge	"frozen" (lowercase)	Finds FROZEN accounts; status "FROZEN"
TC-AS-13	getAccountById	getAccountById_found	Success	id 1 exists	accountType "SAVINGS"
TC-AS-14	getAccountById	getAccountById_notFound	Failure	id 99 does not exist	ResourceNotFoundException "Account not found"
TC-AS-15	getAccountById	getAccountById_noCustomer_customerIdIsNull	Edge	id 1 has no owning customer	customerId is null
TC-AS-16	updateAccount	updateAccount_changesTypeAndBalance	Success	id 1; DTO: CHECKING, ₱250.00	accountType CHECKING, balance 250.00
TC-AS-17	updateAccount	updateAccount_notFound	Failure	id 99 does not exist	ResourceNotFoundException
TC-AS-18	updateAccount	updateAccount_nullStatusKeepsOldStatus	Edge	id 1 is FROZEN; DTO has no status	Status stays FROZEN
TC-AS-19	closeAccount	closeAccount_deletes	Success	id 1 exists	Repository delete called with that account
TC-AS-20	closeAccount	closeAccount_notFound	Failure	id 99 does not exist	ResourceNotFoundException "Account not found"
TC-AS-21	closeAccount	closeAccount_notFound_doesNotDelete	Edge	id 99 does not exist	delete is never called

4.2 CustomerService

Test class: service/CustomerServiceTest.java · Unit (Mockito)

ID	Method	Test method	Type	Input / precondition	Expected result
TC-CS-01	createCustomer	createCustomer_savesAndReturns	Success	New email a@x.com; Alice, 09171234567	fullName "Alice", phone 09171234567
TC-CS-02	createCustomer	createCustomer_duplicateEmail_throws	Failure	Email a@x.com already in use	BusinessRuleException "Email is already in use"
TC-CS-03	createCustomer	createCustomer_duplicateEmail_doesNotSave	Edge	Email a@x.com already in use	save is never called
TC-CS-04	getAllCustomers	getAllCustomers_returnsAll	Success	Repository holds 2 customers	List of 2 customers
TC-CS-05	getAllCustomers	getAllCustomers_empty	Edge	Repository is empty	Empty list
TC-CS-06	getAllCustomers	getAllCustomers_mapsFields	Success	1 customer: Alice, a@x.com	fullName and email mapped
TC-CS-07	getCustomerById	getCustomerById_found	Success	id 1 exists	fullName "Alice"
TC-CS-08	getCustomerById	getCustomerById_notFound	Failure	id 99 does not exist	ResourceNotFoundException "Customer not found"
TC-CS-09	getCustomerById	getCustomerById_setsUsernameWhenLinked	Edge	Customer linked to user "alice"	username "alice"
TC-CS-10	updateCustomer	updateCustomer_changesFields	Success	id 1 (Old, old@x.com); DTO: New, new@x.com, 09171234567	All three fields updated
TC-CS-11	updateCustomer	updateCustomer_notFound	Failure	id 99 does not exist	ResourceNotFoundException
TC-CS-12	updateCustomer	updateCustomer_notFound_doesNotSave	Edge	id 99 does not exist	save is never called
TC-CS-13	deleteCustomer	deleteCustomer_deletes	Success	id 1 exists	Repository delete called with that customer
TC-CS-14	deleteCustomer	deleteCustomer_notFound	Failure	id 99 does not exist	ResourceNotFoundException

ID	Method	Test method	Type	Input / precondition	Expected result
TC-CS-15	deleteCustomer	deleteCustomer_notFound_doesNotDelete	Edge	id 99 does not exist	delete is never called
TC-CS-16	getMyProfile	getMyProfile_found	Success	User "alice" linked to customer Alice	fullName "Alice", username "alice"
TC-CS-17	getMyProfile	getMyProfile_unknownUser	Failure	Username "ghost" does not exist	ResourceNotFoundException "Customer not found"
TC-CS-18	getMyProfile	getMyProfile_userWithoutCustomer	Edge	User "orphan" has no customer profile	ResourceNotFoundException

4.3 TransactionService

Test class: service/TransactionServiceTest.java · Unit (Mockito)

ID	Method	Test method	Type	Input / precondition	Expected result
TC-TS-01	createTransaction	createTransaction_depositAddsToBalance	Success	Account 1 ACTIVE, ₱100.00; DEPOSIT ₱25.50	Balance becomes ₱125.50
TC-TS-02	createTransaction	createTransaction_withdrawalOverBalance_throws	Failure	Balance ₱50.00; WITHDRAWAL ₱75.00	BusinessRuleException "Insufficient funds"; nothing saved
TC-TS-03	createTransaction	createTransaction_transferMovesMoney	Success	Source 1 ₱500.00, target 2 ₱100.00; TRANSFER ₱200.00	Source ₱300.00, target ₱300.00
TC-TS-04	createTransaction	createTransaction_transferToFrozenAccount_throws	Failure	Target 2 is FROZEN; TRANSFER ₱200.00	BusinessRuleException "Account is not active"; source stays ₱500.00
TC-TS-05	createTransaction	createTransaction_depositPastLimit_throws	Edge	BASIC account at ₱9,900.00; DEPOSIT ₱200.00	BusinessRuleException "Wallet limit exceeded for the BASIC tier"
TC-TS-06	getTransactionsByAccount	getTransactionsByAccount_returnsList	Success	Account 1 has 2 transactions	List of 2 transactions
TC-TS-07	getTransactionsByAccount	getTransactionsByAccount_empty	Edge	Account 1 has no transactions	Empty list
TC-TS-08	getTransactionsByAccount	getTransactionsByAccount_mapsType	Success	1 DEPOSIT transaction	type "DEPOSIT"
TC-TS-09	getTransactionById	getTransactionById_found	Success	id 10 exists	type "DEPOSIT"
TC-TS-10	getTransactionById	getTransactionById_notFound	Failure	id 99 does not exist	ResourceNotFoundException "Transaction not found"
TC-TS-11	getTransactionById	getTransactionById_mapsAccountId	Edge	id 10 belongs to account 7	accountId 7
TC-TS-12	updateTransaction	updateTransaction_changesDescription	Success	id 10; description "new note"	description "new note"
TC-TS-13	updateTransaction	updateTransaction_notFound	Failure	id 99 does not exist	ResourceNotFoundException
TC-TS-14	updateTransaction	updateTransaction_ignoresAmountChange	Edge	id 10 (₱100.00); DTO amount ₱999.00	Amount stays ₱100.00

ID	Method	Test method	Type	Input / precondition	Expected result
TC-TS-15	deleteTransaction	deleteTransaction_deletes	Success	id 10 exists	Repository delete called with that transaction
TC-TS-16	deleteTransaction	deleteTransaction_notFound	Failure	id 99 does not exist	ResourceNotFoundException "Transaction not found"
TC-TS-17	deleteTransaction	deleteTransaction_notFound_doesNotDelete	Edge	id 99 does not exist	delete is never called

4.4 AccountController

Test class: controller/AccountControllerTest.java · Slice (@WebMvcTest + MockMvc)

ID	Method	Test method	Type	Input / precondition	Expected result
TC-AC-01	getAccounts	getAccounts_noFilter_returnsAll	Success	GET /api/v1/accounts	200; 2 accounts in data; status filter not used
TC-AC-02	getAccounts	getAccounts_withStatus_filters	Success	GET /api/v1/accounts?status=FROZEN	200; data[0].status FROZEN; full list not used
TC-AC-03	getAccounts	getAccounts_badStatus	Failure	?status=BOGUS (service throws IllegalArgumentException)	400
TC-AC-04	getPaginatedAccounts	getPaginatedAccounts_defaultsTo0And5	Success	GET /api/v1/accounts/page, no parameters	200; service called with page 0, size 5
TC-AC-05	getPaginatedAccounts	getPaginatedAccounts_customPageAndSize	Edge	?page=2&size=10	200; service called with page 2, size 10
TC-AC-06	getPaginatedAccounts	getPaginatedAccounts_nonNumericPage	Failure	?page=abc	400
TC-AC-07	createAccount	createAccount_created	Success	POST /api/v1/accounts: SAVINGS, ₱1,000.00, customerId 1	201; data.id 1
TC-AC-08	createAccount	createAccount_blankType	Failure	accountType ""	400; service not called
TC-AC-09	createAccount	createAccount_customerNotFound	Failure	Valid body; service throws ResourceNotFoundException	404
TC-AC-10	getAccountById	getAccountById_found	Success	GET /api/v1/accounts/1	200; data.id 1
TC-AC-11	getAccountById	getAccountById_notFound	Failure	GET /api/v1/accounts/99	404; message "Account not found"
TC-AC-12	getAccountById	getAccountById_nonNumericId	Edge	GET /api/v1/accounts/abc	400
TC-AC-13	updateAccount	updateAccount_ok	Success	PUT /api/v1/accounts/1, valid body	200; data.status FROZEN

ID	Method	Test method	Type	Input / precondition	Expected result
TC-AC-14	updateAccount	updateAccount_notFound	Failure	PUT /api/v1/accounts/99	404
TC-AC-15	updateAccount	updateAccount_negativeBalance	Edge	balance -5.00	400
TC-AC-16	closeAccount	closeAccount_ok	Success	DELETE /api/v1/accounts/1	200; message "Account closed"; service called with id 1
TC-AC-17	closeAccount	closeAccount_notFound	Failure	DELETE /api/v1/accounts/99	404
TC-AC-18	closeAccount	closeAccount_nonNumericId	Edge	DELETE /api/v1/accounts/abc	400

4.5 AuthController

Test class: controller/AuthControllerTest.java · Slice (@WebMvcTest + MockMvc)

ID	Method	Test method	Type	Input / precondition	Expected result
TC-AU-01	register	register_created	Success	POST /api/v1/auth/register: alice, password123, Alice Cruz, alice@example.com	201 "User registered successfully"; saved with role ROLE_CUSTOMER and an encoded password
TC-AU-02	register	register_duplicateUsername	Failure	Username alice already exists	409 "Username already exists"; nothing saved
TC-AU-03	register	register_duplicateEmail	Failure	Email alice@example.com already exists	409 "Email already exists"; nothing saved
TC-AU-04	register	register_shortPassword	Edge	password "short" (under 8 characters)	400
TC-AU-05	login	login_returnsToken	Success	POST /api/v1/auth/login: alice, password123	200; data is the token
TC-AU-06	login	login_badCredentials	Failure	AuthenticationManager throws BadCredentialsException	401 "Invalid username or password"
TC-AU-07	login	login_blankPassword	Edge	password ""	400; authenticate never called

4.6 CustomerController

Test class: controller/CustomerControllerTest.java · Slice (@WebMvcTest + MockMvc)

ID	Method	Test method	Type	Input / precondition	Expected result
TC-CC-01	createCustomer	createCustomer_created	Success	POST /api/v1/customers: Juan Dela Cruz, juan@example.com, 09171234567	201; message "Customer created"; data.fullName "Juan Dela Cruz"
TC-CC-02	createCustomer	createCustomer_invalidBody	Failure	Blank name, "not-an-email", phone "123"	400
TC-CC-03	createCustomer	createCustomer_duplicateEmail	Failure	Valid body; service throws BusinessRuleException	422 "Email is already in use"
TC-CC-04	getCustomerById	getCustomerById_found	Success	GET /api/v1/customers/1	200; data.id 1
TC-CC-05	getCustomerById	getCustomerById_notFound	Failure	GET /api/v1/customers/99	404 "Customer not found"
TC-CC-06	getCustomerById	getCustomerById_nonNumericId	Edge	GET /api/v1/customers/abc	400
TC-CC-07	getAllCustomers	getAllCustomers_two	Success	GET /api/v1/customers; service returns 2	200; 2 customers in data
TC-CC-08	getAllCustomers	getAllCustomers_empty	Edge	GET /api/v1/customers; service returns none	200; empty data
TC-CC-09	getAllCustomers	getAllCustomers_message	Success	GET /api/v1/customers	message "Listing all customers"
TC-CC-10	updateCustomer	updateCustomer_ok	Success	PUT /api/v1/customers/1, valid body	200; data.fullName "Juan Updated"
TC-CC-11	updateCustomer	updateCustomer_notFound	Failure	PUT /api/v1/customers/99	404
TC-CC-12	updateCustomer	updateCustomer_invalidBody	Edge	PUT with blank name, invalid email and phone	400
TC-CC-13	deleteCustomerById	deleteCustomer_ok	Success	DELETE /api/v1/customers/1	200 "Customer deleted"; service called with id 1
TC-CC-14	deleteCustomerById	deleteCustomer_notFound	Failure	DELETE /api/v1/customers/99	404

ID	Method	Test method	Type	Input / precondition	Expected result
TC-CC-15	deleteCustomerById	deleteCustomer_nonNumericId	Edge	DELETE /api/v1/customers/abc	400
TC-CC-16	getMyProfile	getMyProfile_ok	Success	GET /api/v1/customers/me as alice	200; data.fullName "Alice Cruz"
TC-CC-17	getMyProfile	getMyProfile_notFound	Failure	GET /api/v1/customers/me as ghost (no profile)	404
TC-CC-18	getMyProfile	getMyProfile_usesLoggedInUsername	Edge	GET /api/v1/customers/me as alice	Service called with "alice" from the principal

4.7 TransactionController

Test class: controller/TransactionControllerTest.java · Slice (@WebMvcTest + MockMvc)

ID	Method	Test method	Type	Input / precondition	Expected result
TC-TC-01	createTransaction	createTransaction_created	Success	POST /api/v1/transactions: DEPOSIT ₱100.00, account 1	201; data.id 10
TC-TC-02	createTransaction	createTransaction_zeroAmount	Failure	amount 0	400; service not called
TC-TC-03	createTransaction	createTransaction_businessRule_returns422	Failure	Service throws BusinessRuleException "Insufficient funds"	422 "Insufficient funds"
TC-TC-04	getTransactionsByAccount	getTransactionsByAccount_ok	Success	GET /api/v1/transactions?accountId=1; 2 transactions	200; 2 transactions in data
TC-TC-05	getTransactionsByAccount	getTransactionsByAccount_empty	Edge	?accountId=1; no transactions	200; empty data
TC-TC-06	getTransactionsByAccount	getTransactionsByAccount_missingAccountId	Failure	GET /api/v1/transactions without accountId	400
TC-TC-07	getTransactionById	getTransactionById_found	Success	GET /api/v1/transactions/10	200; data.id 10
TC-TC-08	getTransactionById	getTransactionById_notFound	Failure	GET /api/v1/transactions/99	404 "Transaction not found"
TC-TC-09	getTransactionById	getTransactionById_nonNumericId	Edge	GET /api/v1/transactions/abc	400
TC-TC-10	updateTransaction	updateTransaction_ok	Success	PUT /api/v1/transactions/10, valid body	200
TC-TC-11	updateTransaction	updateTransaction_notFound	Failure	PUT /api/v1/transactions/99	404
TC-TC-12	updateTransaction	updateTransaction_badType	Edge	type "REFUND"	400
TC-TC-13	deleteTransactionById	deleteTransaction_ok	Success	DELETE /api/v1/transactions/10	200 "Transaction deleted"; service called with id 10
TC-TC-14	deleteTransactionById	deleteTransaction_notFound	Failure	DELETE /api/v1/transactions/99	404

ID	Method	Test method	Type	Input / precondition	Expected result
TC-TC-15	deleteTransactionById	deleteTransaction_nonNumericId	Edge	DELETE /api/v1/transactions/abc	400